थाहा नगरपालिका , दामन मकवानपुर
इन्जिनियरिङ सेवा, अधिकृतस्तर छैटौ तह स्त्रस्चरल इन्जिनियरिङ पद
खुला प्रतियोगितात्मक लिखित परीक्षा
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1. **What are the design and construction problems in hill roads? What are special considerations needed to be done in the selection of alignment for roads in high altitude mountainous region? [10]**

Answer:

Hill road can be defined as the road that passes through the area with a cross slope of 25% or more. The hilly regions have extreme climatic condition, rugged topography and high altitude. The roads in these areas are affected by landslides, soil erosion and other problems.

Some of the major design and construction problems associated with hill roads are as follows:

i. **Topographical challenges:**
Highly broken relief, steep slope, deep gorges and large number of watercourses may unnecessarily increase the road length.

ii. **Geological and climatic variations:**
Great variation in the geological, hydrological and climatic conditions from one place to another.

iii. **Slope instability:**
Unstable slope after road construction triggers landslides, rockfalls.

iv. **Installation of various road structures:**
Hill roads require construction of various road structures such as retaining walls, parapet walls, breast walls, side drains, catch-water drains and others.

v. **Design of hairpin bends to gain height:**
Steep elevated area requires hairpin bends and sharp curves which restricts the vehicle speed and other geometric design parameters of hill road.

The special considerations that are needed to be done in the selection of alignment for the roads in high altitude mountainous region are stated below:

i. **Geological Stability:**
- Road alignment should pass through stable hill slopes and should avoid landslide and rockfall prone zones.
- Construction activities might trigger landslide and rockfalls if the road alignment is to be passed through the area having unstable or weak hill slopes.
- Excessive cutting works through hard solid rocks during construction must be avoided.

ii. **Cross drainage structures:**
- Hilly region consists of large number of fast flowing, steep rivers that requires numerous cross-drainage works.
- The road alignment should be selected such that the roadway consists of minimum number of cross drainage structures.

iii. **Geometric Design:**
- Geometric design of hill roads must be done according to the design guidelines and standards for hill region.
- The alignment should have minimum number of sharp curves and hairpin bends, gentler gradient.

iv. **Availability of construction materials:**
- The road alignment should be selected in such a way that the construction materials are available near the construction site.
- It reduces the transportation cost of materials and makes the construction more economical.

v. **Environment and weather:**
- Rainfall (or Snowfall) $\propto$ Altitude
- As the altitude decreases, the number of cross-drainage works required increases.
- At higher altitudes, the road pavement may witness snowfall during winter. This is why the alignment of hill roads should preferably be provided at an altitude between 900 m slopes exposed to sun.

vi. **Length:**
- The cost of construction of a hill road per kilometer length is comparatively very high.
- Hence, it should be ensured that the length of the road connecting two stations should be minimum possible.

2. **Write short notes on:**
a. Role of social mobilization in rural road development.
b. Importance of maintenance in rural road development.

Answer:

a. Role of social mobilization in rural road development:

Social mobilization is the process of organizing and involving local people, community groups, local governments and other stakeholders in identifying, planning, implementing, monitoring and maintaining development activities. It plays an important role in making rural road projects need-based, sustainable, inclusive and locally owned.

The roles of social mobilization in rural development are as follows:

i. **Identification of local needs**
   a. Local people understand their transportation problems and priorities.
   b. Community participation helps select road alignments and locations that provide maximum benefit.

ii. **Community participation**
   a. Mobilizes local people to participate in planning, construction and maintenance.
   b. Encourages a sense of ownership and responsibility toward the road.

iii. **Mobilization of local resources**
   a. Encourages the use of local labor, materials, skills and knowledge.
   b. Reduces construction costs and creates local employment.

iv. **Improved road alignment and design**
   a. Local knowledge helps identify areas vulnerable to landslides, floods, erosion and drainage problems.
   b. Traditional knowledge can complement technical surveys.

v. **Employment generation**
   a. Labor-intensive road construction can provide employment to rural households.
   b. It can particularly benefit disadvantaged and low-income groups.

vi. **Environmental protection**
   a. Communities can help identify environmentally sensitive areas.
   b. Participation can encourage proper drainage, slope protection, bioengineering and controlled disposal of excavated materials.

vii. **Transparency and accountability**
   a. Community monitoring helps control unnecessary expenditure, poor-quality construction and misuse of resources.
   b. Public hearings and social audits can improve transparency

b. Importance of maintenance in rural road development.

Maintenance is the routine and periodic work carried out to keep a rural road in safe, usable and serviceable condition. It is particularly important in Nepal because rural roads are frequently exposed to monsoon rainfall, landslides, floods, erosion and inadequate drainage.

Importance of Maintenance in rural road development:

i. **Extends road service life**
   a. Regular maintenance prevents minor defects from developing into major failures.
   b. It increases the useful life of the road and its structures.

ii. **Reduces total cost**
   a. Preventive maintenance is much cheaper than reconstructing a severely damaged road.
   b. Early repair of potholes, drains and slopes reduces future expenditure.

iii. **Ensures all-season accessibility**
   a. Proper maintenance keeps roads usable during the monsoon and winter.
   b. It ensures access to markets, schools, hospitals and administrative centers.

iv. **Improves road safety**
   a. Maintaining the carriageway, drainage, retaining structures and roadside areas reduces accidents.
   b. Proper signs and removal of obstructions improve safety.

v. **Controls water-related damage**
   a. Water is one of the major causes of rural road deterioration.
   b. Regular cleaning of side drains, cross-drainage structures and culverts prevents water from damaging the road.

vi. **Prevents landslides and erosion**
   a. Timely maintenance of retaining walls, bioengineering measures and drainage systems helps stabilize slopes.
   b. Removal of loose material and debris reduces blockage and further slope failure.

vii. **Supports rural economic development**
   a. Reliable roads reduce transportation costs and travel time.
   b. Farmers can transport agricultural products to markets more easily, while businesses and tourism benefit from improved connectivity.

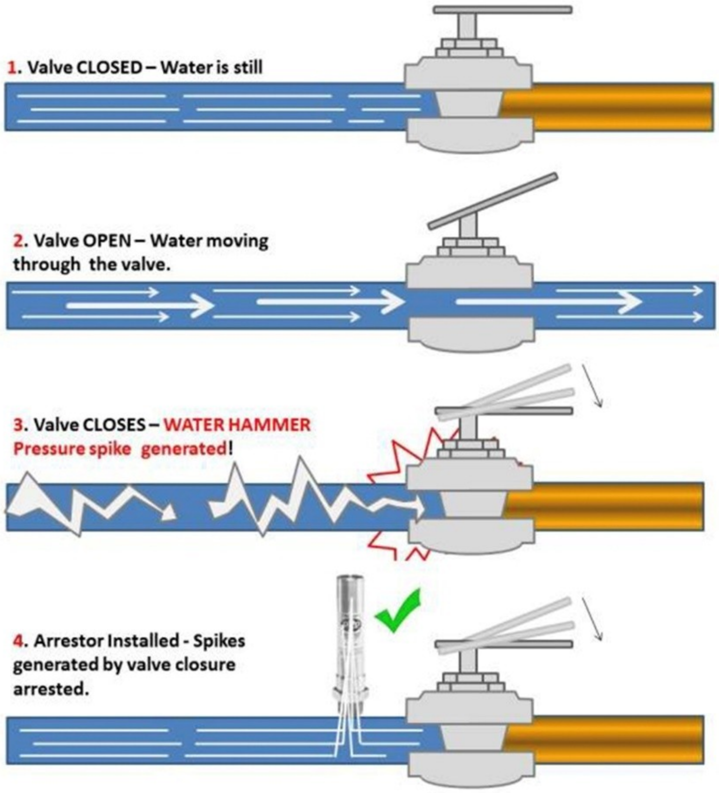
viii. **Improves emergency response**
   a. Well-maintained roads provide reliable access for ambulances, fire services, disaster response teams and relief materials, especially during floods and earthquakes.

ix. **Promotes local employment**
   a. Routine maintenance can be carried out using local labor and locally available resources, creating employment in rural communities.

3. **What do you understand by water hammer? Explain different types of surge tanks with scheme drawings. [10]**

Answer:

If a liquid flowing in a pipe is suddenly brought to rest by closing the valve, there is an abrupt rise in the pressure because the momentum of liquid is destroyed. A pressure wave is transmitted along the pipe. The velocity of this pressure wave is equal to the velocity of sound in the water. The pressure wave and the associated pulsation or fluctuations in the pressure produce the noise known as water hammer or knocking.



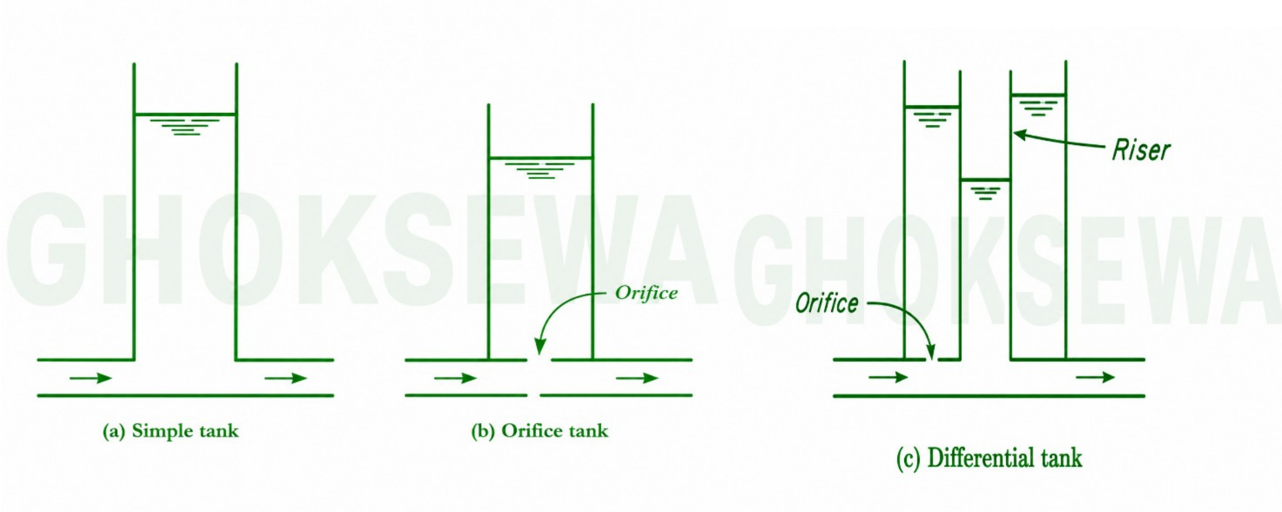
Surge tanks are provided to reduce the water hammer or elastic shock wave coming from the pressure pipeline or tunnel on closing and opening action of the valve.

Surge tank is the cylindrical storage tank which is connected to the penstock at suitable point. It is provided in the case of hydropower projects having long penstock pipes. Surge tank is divided into:

i. Simple surge tank:
   - It consists of a shaft connected to pressure tunnel directly or by a short connection of cross-sectional area mot less than the area of the headrace tunnel.
   - It acts as a reservoir directly connected to the penstock line in such a manner that the water flows into and out of the tank without any head loss.

ii. Restricted orifice or throttled surge tank:
   - It is a surge tank in which inlet is throttled to improve damping of oscillation by offering greater resistance and connected to the headrace tunnel with or without a connecting shaft.
   - It decreases the amplitude of the surge and effects more rapid reduction if the mass oscillation and gives economy up to 40% in comparison to the simple tank.

iii. Differential surge tank:
   - It consists of a combination of the simple surge tank and the restricted orifice surge tank except that on internal riser pipe extends upward from the restricted orifice to near the top of the outside shell.
   - It separates out the two functions of simple storing of additional water and effective accelerating and decelerating of water in the main conduit by outside main tank and central river respectively.



4. **Has technology brought changes in education and employment opportunity of Nepal? Discuss. [10]**

Answer:

Yes. Technology has brought significant changes in both education and employment opportunities in Nepal**.** The expansion of internet access, smartphones, digital platforms, online learning, information and communication technology (ICT), automation and artificial intelligence has changed how people learn, work and access economic opportunities. At the same time, unequal digital access and skill gaps remain major challenges.

a. Online and digital learning:

Students can now access lectures, e-books, videos, online courses and educational platforms from home. This became particularly important during the COVID-19 pandemic and has continued to expand.

b. Wider access to information:

The internet has made educational resources available beyond textbooks and physical libraries. Students can access international journals, tutorials, simulations and open educational resources.

c. Improved teaching methods:

Teachers can use multimedia presentations, digital boards, educational applications, virtual laboratories and online assessments to make learning more interactive.

d. Distance and flexible education:

Technology has enabled students in remote areas and working students to participate in online classes and training without always having to travel to major cities.

e. Development of digital skills:

Students are increasingly learning computer programming, digital design, data analysis, communication technologies and other skills relevant to modern employment. Recent discussions among Nepali youth also show strong interest in digital skills, entrepreneurship, creative industries and technology-related careers.

Technology has created new types of employment as well as changed traditional occupations.

a. Software development and IT services
b. Web and mobile application development
c. Graphic design and digital marketing
d. Content creation and social media management
e. Data analysis and AI-related work
f. Online teaching and tutoring
g. Freelancing and remote employment

h. E-commerce and digital entrepreneurship

Digital platforms have also made it possible for Nepali workers to provide services to international clients without physically migrating abroad. The digital economy includes areas such as ICT services, BPO, e-commerce, digital financial services and online employment platforms. Technology is also encouraging entrepreneurship. For example, recent World Bank-supported programs in Nepal have provided students and graduates with entrepreneurship training, mentoring and seed financing, helping them turn ideas into businesses and employment.

The following measures can be adopted to maximize the benefits of technology on education and employment:

a. Expand affordable high-speed internet to rural and remote areas.
b. Improve electricity and digital infrastructure in schools.
c. Train teachers in digital pedagogy.
d. Integrate digital, technical and vocational skills into curricula.
e. Promote STEM, AI, coding, data science and entrepreneurship.
f. Strengthen industry–university collaboration.
g. Expand online job and freelancing opportunities.
h. Provide digital-skills training to women and disadvantaged communities.
i. Establish stronger cybersecurity and data-protection systems.
j. Promote lifelong learning and reskilling as technology changes employment.

5. **What is the difference between renewable and non-renewable energy? Also write down advantages of solar energy, bio-gas and hydropower.**

Answer:

Renewable energy is clean, sustainable, and environmentally friendly, making it the preferred option for future energy development. Non-renewable energy provides reliable power but is finite, polluting, and unsustainable. The differences between renewable and non-renewable energy are mentioned below:

Renewable energy	Non-renewable energy
i. Energy obtained from the natural sources that are replenished continuously is known as renewable energy.	i. Energy obtained from finite natural resources that cannot be replenished within a human lifetime is known as non-renewable energy.
ii. It is sustainable source of energy and can be used repeatedly.	ii. It is exhaustible source of energy and will eventually run out.
iii. It produces little to no pollution and greenhouse gasses.	iii. it produces significant pollution and greenhouse gas emissions.
iv. It is environmentally friendly.	iv. it causes environmental degradation and contributes to climate change.
v. Renewable energy has high installation cost but low operating cost.	v. Non-renewable energy has lower initial cost but higher fuel and operating costs over time.
vi. It is suitable for long-term energy security.	vi. it is not sustainable for long-term energy needs.
vii. Example: solar, wind, hydropower	vii. Example: coal, petroleum, nuclear fuel etc.

i. Solar energy:

Solar energy provides about 7% of the world’s electricity supply but only 2 to 3% of total global primary energy consumption.

Advantages

- Renewable and inexhaustible source of energy.
- Environment-friendly with no air pollution during operation.

- Suitable for remote and rural areas where grid electricity is unavailable.
- Low operation and maintenance costs.
- Reduces dependence on fossil fuels.

Disadvantages

- High initial installation cost.
- Electricity generation depends on sunlight and weather conditions.
- Requires large land area for large-scale solar plants.
- Energy storage (batteries) increases system cost.
- Lower efficiency during cloudy or rainy seasons.

ii. Biogas:

Biogas fulfills less than 1% of the total global energy demand.

Advantages

- Renewable source produced from organic wastes such as animal dung and kitchen waste.
- Provides clean fuel for cooking and lighting.
- Reduces indoor air pollution and firewood consumption.
- Produces nutrient-rich slurry that can be used as organic fertilizer.
- Helps in proper waste management and sanitation.

Disadvantages

- Requires a regular supply of organic waste and water.
- Gas production decreases during cold weather.
- High initial construction cost for the digester.
- Requires proper maintenance and technical knowledge.
- Not suitable where sufficient organic waste is unavailable.

iii. Hydropower:

Hydropower fulfills 14-15% of the global electricity demand.

Advantages

- Renewable and clean source of energy.
- Low operating and maintenance costs after construction.
- High efficiency and long service life.
- Produces very low greenhouse gas emissions.
- Nepal has abundant hydropower potential due to its rivers and steep terrain.

Disadvantages

- Very high initial construction cost.
- Construction period is long.
- Can affect aquatic ecosystems and local communities.
- Power generation depends on river flow, which may reduce during the dry season.
- Large projects may require relocation of people and environmental mitigation measures.

6. **Describe various types of river training and protection works.** **[10]**

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

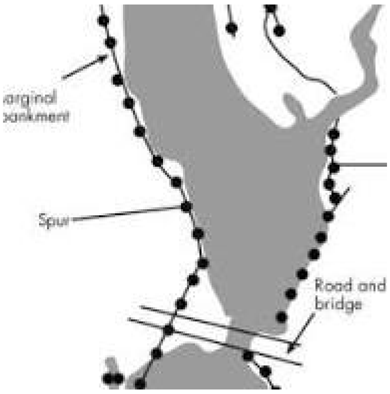
In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

Depending upon the purpose required river training works can be classified into three types.

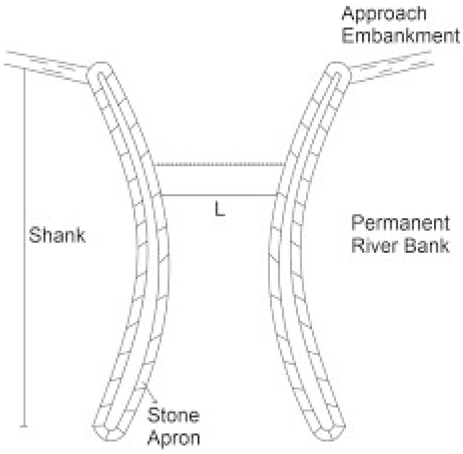
i. high water training works: training for discharge
ii. low water training works: training for depth
iii. medium water training works: training for sediment

There are different methods of river training works. Some of them are described below:

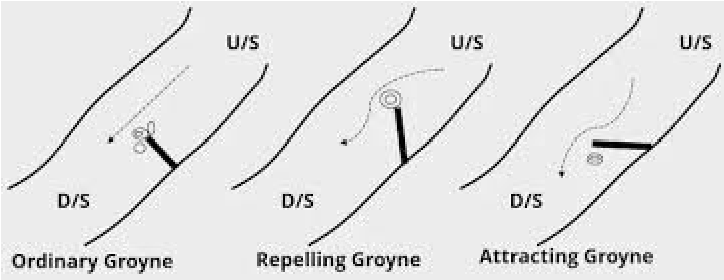
i. **Marginal embankments/ levees:**
   - Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
   - They are usually placed at some distance away from the riverbank so that area of the flow is increased.
   - Generally required for meandering rivers.



ii. **Guide banks:**
   - Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
   - The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.

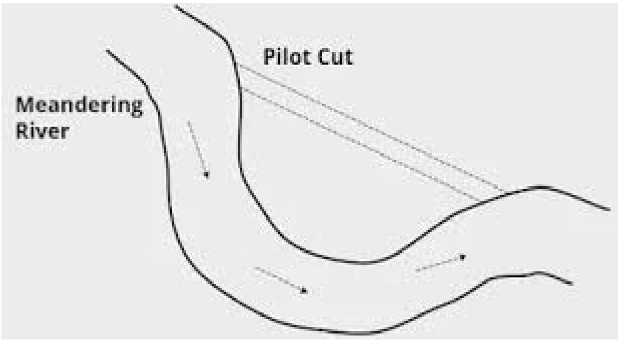


iii. **Groynes or spurs:**
   - Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
   - They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. **Artificial cutoff:**

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.



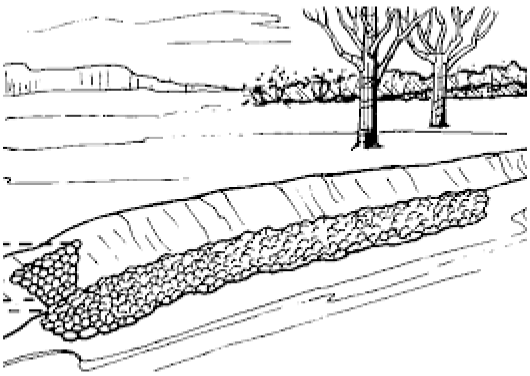
v. **Bank protection and Pitched banks:**

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

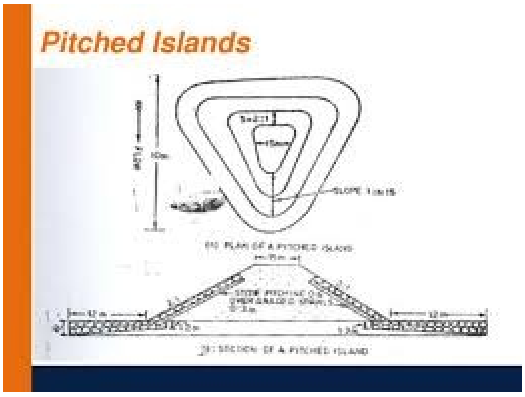
Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



vi. **Pitched islands:**

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. **Bandalling:**

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.

7. **What are the roles of National Building Codes in Nepal? How does the code address the problem of earthquake? How could the code be made effective? [10]**

Answer:

Nepal National Building Code (NBC) is a set of minimum technical requirements and guidelines for the safe planning, design and construction of buildings in Nepal. It provides standards for structural safety, earthquake resistance, loads, materials, fire safety and other building requirements. The Department of Urban Development and Building Construction (DUDBC) currently publishes the NBC series, including NBC 105: 2025 – Seismic Design of Buildings in Nepal and guidelines for masonry, earthen and reinforced-concrete buildings.

Roles of National Building Codes in Nepal:

The major roles are:

i. **Ensuring structural safety**
Establishes minimum requirements for the strength, stability and serviceability of buildings.

ii. **Controlling building design and construction**
Provides standardized requirements for structural systems, materials, foundations, loads and construction practices.

iii. **Reducing disaster risk**
Addresses major hazards such as earthquakes, wind, snow and fire through specific codes and guidelines.

iv. **Protecting life and property**
Proper compliance reduces the probability of structural failure and loss of life during disasters.

v. **Supporting building approval and regulation**
Provides technical criteria against which municipalities can review building designs and construction.

Nepal is highly vulnerable to earthquakes, so seismic safety is one of the most important aspects of the NBC.

a. Seismic design requirements: NBC 105 – Seismic Design of Buildings in Nepal provides procedures and criteria for seismic analysis and design. It applies to buildings constructed using reinforced concrete, steel, composite, timber and masonry systems.

b. Earthquake forces

The code specifies procedures for determining the minimum design earthquake forces acting on buildings and provides methods for seismic analysis and design.

c. Seismic hazard considerations

The code considers Nepal's seismicity and provides appropriate seismic design parameters for structural design. Site-related seismic hazards are also addressed separately through NBC 108: Site Consideration for Seismic Hazards.

d. Performance-based considerations

The revised seismic provisions establish performance requirements related to damage limitation and collapse prevention and provide procedures for structural analysis.

e. Earthquake-resistant construction for different building types
Nepal also has specific guidelines such as:

- **NBC 201** – Mandatory Rules of Thumb for RCC buildings with masonry infill
- **NBC 203** – Earthquake-resistant construction of low-strength masonry
- **NBC 204** – Earthquake-resistant construction of earthen buildings
- **NBC 205** – Detailing guidelines for low-rise RCC buildings

Thus, the NBC addresses earthquakes not only through structural calculations but also through site selection, structural configuration, detailing, construction practices and retrofitting**.**

The major challenge in Nepal is not only the availability of codes but their effective implementation and compliance. Studies following the 2015 Gorkha earthquake identified irregular compliance, owner-built construction, inadequate training and differences between approved designs and actual construction as important problems.

The following measures can improve effectiveness:

i. Strengthen enforcement

  - Municipalities should conduct regular site inspections.
  - Construction should be checked at critical stages such as foundation, reinforcement and structural framing.
  - Completion certificates should be issued only after proper verification.

ii. Improve capacity of local governments

- Municipalities need sufficient numbers of qualified engineers, sub-engineers and building inspectors to review designs and monitor construction.
- Strengthening design checking and inspection capacity is also identified as an important component of improving NBC compliance.

iii. Train masons and contractors

- Since a large amount of construction is owner-built or locally supervised, practical training should be provided to:

    a. Masons
    b. Contractors

c. Engineers
d. Architects
e. Sub-engineers
f. Homeowners

- Training should focus particularly on earthquake-resistant detailing and construction quality.

Building codes should be periodically revised using new research, lessons from earthquakes and advances in engineering technology. Nepal has already revised its seismic code, with DUDBC listing NBC 105: 2025 as the current seismic-design standard.

8. **Write short notes on:**
a. Farmers managed irrigation system
b. Specific consideration in design of buildings in Nepal

Answer:

a. Farmer Managed Irrigation Systems (FMIS)

- **Farmer Managed Irrigation Systems (FMIS)** are irrigation schemes that are developed, operated, and maintained entirely by the farming communities they serve. In the context of Nepal, these systems represent a significant part of the agricultural landscape and have a long historical tradition.
- Before 1922, irrigation in Nepal was primarily developed and managed through FMIS. It was only after this period that the government began making significant efforts to develop state-led irrigation infrastructure.
- FMIS are the backbone of Nepal's irrigated agriculture, covering approximately **70% of the country's total irrigated land**.
- Some of the biggest FMIS maintained in Nepal are Rajapur Irrigation System, Rani Jamara Kulariya Irrigation System etc.
- Advantages of FMIS**:**
  i. Strong community participation and ownership.
  ii. Low operation and maintenance cost due to farmers' contributions.
  iii. Efficient and equitable water distribution based on local rules.
  iv. Quick repair and maintenance because users are directly involved.
  v. Uses local knowledge and indigenous technology.
  vi. Improves agricultural productivity and rural livelihoods.
  vii. Promotes social cooperation and conflict resolution among farmers.
  viii. Sustainable due to users' sense of responsibility.

b. Specific Consideration of Design of bridge in Nepal:

Specific design considerations to avoid bridge failures are described in brief:

i. **Determination of load:**
Road bridges are subjected to heavy vehicular traffic in addition to their own self-weight. Therefore, they must be designed to safely withstand dead loads, live loads, impact loads, and other applicable forces throughout their service life.

ii. **Hydraulic and geotechnical safety:**
Road bridges must be designed to withstand extreme hydrological, geological and climatic conditions to avoid failure of bridges due to extreme flood, landslides and scour.

iii. **Ductility:**
Redundancy and ductility of bridge should be improved by proper detailing so that when one component of structure fails, it can be supported by another component.

9. **Briefly explain the following:** **[5+5=10]**

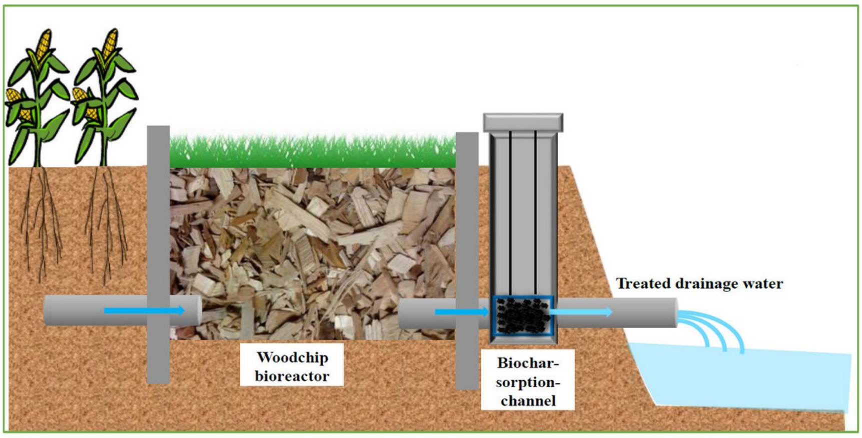
a. Remedial measures of water logging
b. Indigenous technology in building designs
Answer:
a. Remedial measures of water logging:
Water logging is the condition of saturation of soil such that is air phase is restricted and anaerobic conditions prevails. Waterlogging is often accompanied by soil salinity as waterlogged soils prevent leaching of the salts imported by the irrigation water. Due to water logging, groundwater table rises close to the ground surface (generally within **1.5–2.0 m**), causing the root zone to remain saturated and reducing soil aeration.

- Water logging is an increasing problem in flat terrains of terai region and low-lying pocket lands in Nepal.
- Water-logging is caused by intensive irrigation, seepage from canals/ reservoirs, seepage from adjoining lands, lack of adequate drainage systems, monsoon rainfall and so on.
- Water logging has caused hamper in nitrification which reduces the crop yield and also leads to rise of salt in surface layer and overgrowth of weeds.

The remedial measures of water logging are as follows:

i. **Surface drainage system:**
   - The removal of excess rainwater falling on the fields or excess irrigation water applied to the fields by constructing open ditches. Field drains and other structures is known as surface drainage system.
   - The land is loped toward the ditches or drains making excess flow of water to remove from the drains.
   - Surface drains can be divided into surface drain and deep surface drains.

ii. **Subsurface drainage system:**
   - The removal of excess rainwater by using buried perforated pipes or tile drains to lower the groundwater table is known as subsurface drainage system.
   - It is most effective for controlling water logging in agricultural fields.



iii. **Interceptor drains:**
   - These drains are constructed along the canals or hill slope to intercept seepage before it reaches agricultural land.

iv. **Bio-drainage:**
   - Plantation of deeply rooted trees, highly water absorbing plants to lower the groundwater table naturally.

v. **Open drainage system:**
   - It consists of open ditches that carry water to nearby rivers or canals.
   - It is simple and economical for large agricultural areas.

b. Indigenous technologies in building design:

Indigenous building technologies are traditional construction methods, materials, knowledge and practices developed and adapted by local communities over generations according to their climate, geography, available resources, culture and local needs. In Nepal, these technologies provide valuable lessons for sustainable, economical and climate-responsive building design.

Major Indigenous Technologies in Nepal

i. **Traditional stone masonry**
   a. Use of locally available natural stone for walls and foundations.
   b. Common in hilly and mountainous regions.
   c. Provides good thermal mass and durability when properly constructed.

ii. **Mud and adobe construction**
   a. Use of locally available soil/mud for walls and adobe blocks.
   b. Provides good thermal insulation.
   c. Traditionally common in rural settlements.

iii. **Timber construction**
   a. Local timber is used for beams, columns, floors, roofs and structural framing.
   b. Traditional systems such as Kathmandu Valley's Newari timber-framed buildings.
   c. demonstrate sophisticated use of wood.

iv. **Dhungaa–Mato (stone-mud) houses**
   a. Stone masonry bonded with mud mortar.
   b. Economical and uses locally available materials.
   c. Requires appropriate detailing and strengthening for seismic safety.

v. **Thatch and traditional roofing**
   a. Grass, straw, slate, tiles and locally available materials have traditionally been used for roofs.
   b. Suitable materials can provide good thermal performance.

vi. **Traditional brick and tile construction**
   a. Fired bricks and clay tiles are widely used in the Kathmandu Valley and other parts of Nepal.
   b. Traditional brickwork also has important cultural and architectural value.

vii. **Courtyard planning**
   a. Buildings are arranged around courtyards to provide light, ventilation, social interaction and climatic comfort.
   b. A prominent feature of traditional Newari settlements.

Advantages:

i. Uses locally available materials and skills.
ii. Reduces transportation and construction costs.
iii. Creates local employment.
iv. Has relatively low environmental impact when sustainably managed.
v. Provides climate-responsive thermal performance.

vi. Preserves Nepal's cultural heritage.
vii. Reduces dependence on imported construction materials.

10. **Write short notes on: [5+5=10]**
a. Initial Environment Examination
b. Labor based, Environment friendly and participatory approach for local infrastructure development in Nepal.

Answer:

a. Initial Environment Examination:

- According to Environment Protection Act (EPA) 2076 and Environment Protection Rules (EPR) 2077, IEE is defined as "An analytical study or evaluation conducted to determine whether a proposed project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts."
- Done for medium scale projects that has moderate effect on environment.
- Project requiring IEE are listed in Schedule 2 of EPR 2077.
- Includes:

  Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → IEE Report Preparation → Recommendation and submission → final review and approval → Decision making
- If IEE Report submitted is approved by concerned authority, the project is implemented.
- IEE is done for:
  a. Highways with the length of road 5-50 km
  b. Irrigation project with command area 25-2000 hectares.
  c. Hydropower project with installed capacity of 1 MW-50 MW

b. Labor based, environmentally friendly and participatory approach for local infrastructure development in Nepal:

A labor-based, environmentally friendly and participatory approach is a method of developing local infrastructure in which local labor, locally available materials, community participation and environmentally sustainable practices are prioritized instead of relying mainly on heavy machinery and external resources. It is particularly suitable for rural roads, trails, irrigation canals, drainage, small bridges, water supply and community buildings in Nepal.

Labor-based approach
The labor-based approach emphasizes the use of local human resources and appropriate levels of machinery rather than complete mechanization. It promotes the labor-intensive technologies such as stone masonry, gabion works, dry-stone structures and manual road construction. It reduces dependence on expensive machinery, creates employment, retains money within the local economy and can be particularly effective in remote areas where transporting heavy equipment is difficult.

Environmentally friendly approach
Infrastructure should be planned and constructed with minimum adverse effects on the land, forests, water resources and local ecosystems. The approach incorporates the climate-resilient designs considering floods, landslides and changing rainfall patterns. It also provides proper operation and maintenance to increase infrastructure life and reduce repeated resource consumption.

Participatory approach
The local community should be involved throughout the identification, planning, implementation, monitoring and maintenance of infrastructure projects. The community can help identify actual needs, select project locations, provide local knowledge and participate in construction and maintenance.